The Yield Recurrence

May 13, 2026

The recurrence we consider in this section is the following.

$$Y_{n+2} \stackrel{d}{=} I^{(n)} \left(Y^{(1)}_{U_n} + Y^{(2)}_{n+1-U_n} + 1 \right) + J^{(n)} \max \left(Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n} \right), \forall n \geq 1, \quad Y_1 = Y_2 = 0. \tag{1.1}$$

Let us explain the notations. Let Y_n be the yield of the tree with n nodes. The variables I and J are Bernoulli with expectation u and v respectively, such that $I + J \leq 1$. That means we can only either take the sum-plus-one term, or the max term, or zero. The variable U_n is uniformly distributed on $\{1, \dots, n\}$. Superscripts are for independent copies. We assume that all collections of copies are independent. Now we can understand why it is correct to reuse the same variables $Y^{(1)}_{U_n}, Y^{(2)}_{n+1-U_n}$ and U_n across both terms.

1.1 Lower and Upper Bounds of the Expectation

Let $y_n = \mathbb{E}[Y_n]$. Take expectation on both sides, we have

$$y_{n+2} = u + \frac{2u}{n} \sum_{i=1}^{n} y_i + \frac{v}{n} \sum_{i=1}^{n} \mathbb{E}[\max(Y_i, Y_{n+1-i})].$$

For an upper bound, using $\max(Y_i, Y_{n+1-i}) \leq Y_i + Y_{n+1-i}$, we get an upper bound sequence

$$y_{n+2} = u + \frac{2(u+v)}{n} \sum_{i=1}^{n} y_i, \quad y_1 = y_2 = 0. \tag{1.2}$$

There are several ways to get a lower bound. The first one is to use convexity of the absolute value function.

$$\begin{aligned}
\mathbb{E}[\max(Y_i, Y_{n+1-i})] &= \mathbb{E}\left[\frac{1}{2}(Y_i + Y_{n+1-i} + |Y_i - Y_{n+1-i}|) \right] \\
&\geq \frac{1}{2}(\mathbb{E}[Y_i] + \mathbb{E}[Y_{n+1-i}]) + \frac{1}{2}|\mathbb{E}[Y_i] - \mathbb{E}[Y_{n+1-i}]| \\
&= \frac{1}{2}(y_i + y_{n+1-i} + |y_i - y_{n+1-i}|).
\end{aligned}$$

This gives

$$y_{n+2} = u + \frac{2u+v}{n} \sum_{i=1}^{n} y_i + \frac{v}{2n} \sum_{i=1}^{n} |y_i - y_{n+1-i}|, \quad y_1 = y_2 = 0. \tag{1.3}$$

Another way is to use convexity of the max function.

$$\mathbb{E}[\max(Y_i, Y_{n+1-i})] \geq \max(\mathbb{E}[Y_i], \mathbb{E}[Y_{n+1-i}]) = \max(y_i, y_{n+1-i}) = y_{\max(i, n+1-i)}.$$

This gives

$$y_{n+2} = u + \frac{2u}{n}\sum_{i=1}^{n} y_i + \frac{v}{n}\sum_{i=1}^{n} y_{\max(i,n+1-i)}, \quad y_1 = y_2 = 0. \tag{1.4}$$

If we can prove that y_n is non-decreasing in n, then we can ignore the middle term in the sum of maxima, and get

$$y_{n+2} = u + \frac{2u}{n}\sum_{i=1}^{n} y_i + \frac{2v}{n}\sum_{i=\lfloor (n+1)/2\rfloor}^{n} y_i, \quad y_1 = y_2 = 0. \tag{1.5}$$

The weakest lower bound is to replace the max by the average, which gives

$$y_{n+2} = u + \frac{2u+v}{n}\sum_{i=1}^{n} y_i, \quad y_1 = y_2 = 0. \tag{1.6}$$

When $2u+v>1$, the true expectation is bounded between by polynomial of degrees $2u+v-1$ and $2(u+v)-1$. This case agrees with real data. We conjecture that the true expectation also grows polynomially. Details of the analysis of the deterministic recurrences are in Chapter **??**.

1.2 Combinatorial Analysis

For convenience, let us shift the index. The recurrence becomes

$$Y_{n+1} \stackrel{d}{=} I^{(n)}\left(Y^{(1)}_{U_n} + Y^{(2)}_{n-1-U_n} + 1\right) + J^{(n)}\max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n-1-U_n}\right), \forall n \geq 1, \quad Y_0 = Y_1 = 0. \tag{1.7}$$

The variable U_n is now distributed on $\{0,\ldots,n-1\}$. Let $p_{n,m} = \mathbb{P}(Y_n = m)$ and $q_{n,m} = \mathbb{P}(Y_n \leq m)$. We have the relations

$$p_{n,m} = q_{n,m} - q_{n,m-1} \text{ and } q_{n,m} = \sum_{i=0}^{n-1} p_{i,m},$$

and initial conditions

$$p_{0,0} = p_{1,0} = 1 \text{ and } q_{0,m} = q_{1,m} = 1 \text{ for all } m \geq 0.$$

From the recurrence, we have

$$\begin{aligned}
&nq_{n+1,m}\\
&= nu\mathbb{P}\left(Y^{(1)}_{U_n} + Y^{(2)}_{n-1-U_n} + 1 \leq m\right) + nv\mathbb{P}\left(\max\left(Y^{(1)}_{U_n}, Y^{(2)}_{n-1-U_n}\right) \leq m\right) + n(1-u-v)\\
&= u\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} p_{i,j}q_{n-1-i,m-1-j} + v\sum_{i=0}^{n-1} q_{i,m}q_{n-1-i,m} + n(1-u-v)\\
&= u\left(\sum_{i=0}^{n-1}\sum_{j=0}^{m-1} q_{i,j}q_{n-1-i,m-1-j} - \sum_{i=0}^{n-1}\sum_{j=0}^{m-2} q_{i,j}q_{n-1-i,m-2-j}\right)\\
&\quad + v\sum_{i=0}^{n-1} q_{i,m}q_{n-1-i,m} + n(1-u-v).
\end{aligned}$$

Let $Q_m(z) = \sum_{n=0}^{\infty} q_{n,m} z^n$ be the generating function of $q_{n,m}$ when m is fixed. Multiplying both sides by z^{n-1} and summing over n, we get

$$\sum_{n=0}^{\infty} n\, q_{n+1,m} z^{n-1} = \frac{d}{dz} \sum_{n=0}^{\infty} q_{n+1,m} z^n = \frac{d}{dz}\left(\frac{Q_m(z)-1}{z}\right) = \frac{zQ'_m(z) - Q_m(z) + 1}{z^2},$$

$$\sum_{n=0}^{\infty} \sum_{i=0}^{n-1} \sum_{j=0}^{m-1} q_{i,j} q_{n-1-i,m-1-j} z^{n-1} = \sum_{j=0}^{m-1} Q_j(z) Q_{m-1-j}(z),$$

$$\sum_{n=0}^{\infty} \sum_{i=0}^{n-1} \sum_{j=0}^{m-2} q_{i,j} q_{n-1-i,m-2-j} z^{n-1} = \sum_{j=0}^{m-2} Q_j(z) Q_{m-2-j}(z),$$

$$\sum_{n=0}^{\infty} \sum_{i=0}^{n-1} q_{i,m} q_{n-1-i,m} z^{n-1} = Q_m(z)^2,$$

$$\sum_{n=1}^{\infty} n(1-u-v) z^{n-1} = \frac{1-u-v}{(1-z)^2}.$$

We get the following functional equation for $Q_m(z)$.

$$zQ'_m(z) - Q_m(z) + 1 = z^2 \left(u \sum_{j=0}^{m-1} Q_j(z) Q_{m-1-j}(z) - u \sum_{j=0}^{m-2} Q_j(z) Q_{m-2-j}(z) + vQ_m(z)^2 \right) + \frac{(1-u-v)z^2}{(1-z)^2}. \tag{1.8}$$

In particular, when $m = 0$, we have

$$zQ'_0(z) - Q_0(z) + 1 = vz^2 Q_0(z)^2 + \frac{(1-u-v)z^2}{(1-z)^2}. \tag{1.9}$$

By the convention of Riccati equations, let $Q_0(z) = -\frac{1}{vz}\frac{W'(z)}{W(z)}$, we have

$$z\left(\frac{1}{vz^2}\frac{W'(z)}{W(z)} - \frac{1}{vz}\frac{W''(z)W(z) - W'(z)^2}{W(z)^2}\right) + \frac{1}{vz}\frac{W'(z)}{W(z)} + 1 = \frac{1}{v}\frac{W'(z)^2}{W(z)^2} + \frac{(1-u-v)z^2}{(1-z)^2}.$$

Equivalently,

$$W''(z) - \frac{2}{z}W'(z) - v\left(1 - \frac{(1-u-v)z^2}{(1-z)^2}\right) W(z) = 0. \tag{1.10}$$

Our approach follows closely [1].

Step 1. By Cauchy-Hadamard theorem, the radius of convergence of $Q_0(z)$ is larger than or equal to 1. Suppose that it is strictly larger than 1. Then $Q_0(z), Q'_0(z)$ is finite. Taking the limit of both sides of (1.9) as $z \to 1^-$, we get a contradiction. Therefore, the radius of convergence of $Q_0(z)$ is exactly 1.

Step 2. We find the indicial equation of the ODE at $z = 1$.

$$\delta_1 = \lim_{z\to 1}(z-1)\left(-\frac{2}{z}\right) = 0,$$

$$\delta_2 = \lim_{z\to 1} -(z-1)^2 v\left(1-\frac{(1-u-v)z^2}{(1-z)^2}\right) = \lim_{z\to 1} -v\left((z-1)^2-(1-u-v)z^2\right) = v(1-u-v).$$

Solving $I(\theta) = \theta(\theta-1)+\delta_1\theta+\delta_2 = 0$, we get

$$\theta_{\pm} = \frac{1\pm\sqrt{1-4v(1-u-v)}}{2}.$$

We have $\theta_+ - \theta_- = \sqrt{1-4v(1-u-v)} \in [0,1]$. Choose the star-shaped domain $\tilde{\Omega} = (\mathbb{C}\setminus[1,\infty))\cap D(3/4,1/2)$. There are two cases for the solution of the ODE in $\tilde{\Omega}$ near $z=1$.

1. If $v(1-u-v) < \dfrac{1}{4}$, then

$$W(z) = c_1(1-z)^{\theta_+}H_1(1-z) + c_2(1-z)^{\theta_-}H_2(1-z).$$

2. If $v(1-u-v) = \dfrac{1}{4}$, then $u=0$ and $v=\dfrac{1}{2}$. Hence,

$$W(z) = (1-z)^{1/2}\left(c_1 H_1(1-z) + c_2\log(1-z)H_2(1-z)\right).$$

Step 3. We pass the asymptotic information of $W(z)$ to $Q_0(z)$.

Another Upper Bound on the Yield Recurrence

May 20, 2026

Let us recall the yield recurrence

$$Y_{n+2} \overset{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)} \right), \forall n \geq 1, \quad Y_1 = Y_2 = 0.$$

2.1 Escaping from the Random Variables

Let $Z_n = \lambda^{Y_n}$, where $\lambda > 0$ is a constant to be determined later. For every $I \in \{0, 1\}$ and $C \in \mathbb{R}$, we have

$$C^I = I \cdot C + (1 - I).$$

Since $I^{(n)} J^{(n)} = 0$, we have

$$\begin{aligned}
&\lambda^{Y_{n+2}} \\
&= \left(I^{(n)} \left(\lambda^{Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1} \right) + 1 - I^{(n)} \right) \left(J^{(n)} \left(\lambda^{\max(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)})} \right) + 1 - J^{(n)} \right) \\
&= I^{(n)} \left(\lambda^{Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1} \right) + J^{(n)} \lambda^{\max\left(Y_{U_n}^{(1)}, Y_{n+1-U_n}^{(2)}\right)} + 1 - J^{(n)} - I^{(n)} \\
&\leq I^{(n)} \left(\lambda^{Y_{U_n}^{(1)} + Y_{n+1-U_n}^{(2)} + 1} \right) + J^{(n)} \left(\lambda^{Y_{U_n}^{(1)}} + \lambda^{Y_{n+1-U_n}^{(2)}} \right) + 1 - I^{(n)} - J^{(n)}
\end{aligned}$$

Let $z_n = \mathbb{E}[Z_n]$. Taking the expectation, we get

$$z_{n+2} \leq \frac{\lambda u}{n} \sum_{i=1}^{n} z_i z_{n+1-i} + \frac{2v}{n} \sum_{i=1}^{n} z_i + 1 - u - v.$$

Hence let us consider the sequence

$$n z_{n+2} = \lambda u \sum_{i=1}^{n} z_i z_{n+1-i} + 2v \sum_{i=1}^{n} z_i + n(1 - u - v), \forall n \geq 1, \quad z_1 = z_2 = 1. \tag{2.1}$$

Experiments show that (z_n) grows exponentially, the bound fails. That means, using $\max(x, y) \leq x + y$ keeps the sequence sublinear, but using $\lambda^{\max(x,y)} \leq \lambda^x + \lambda^y$ pushes it to linear.

2.2 Using Techniques from Analytic Combinatorics

Let $Z(t) = \sum_{n=1}^{\infty} z_n t^n$ be the generating function for the sequence $\{z_n\}$. Let us analyze each term in (2.1) through Z.

$$\begin{aligned}
&\sum_{n=1}^{\infty} n z_{n+2} t^n = t\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Z(t) - z_1 t - z_2 t^2}{t^2}\right) = t\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{Z(t)}{t^2} - \frac{1}{t} - 1\right) = \frac{Z'(t)}{t} - \frac{2Z(t)}{t^2} + \frac{1}{t},\\
&\sum_{n=1}^{\infty}\sum_{i=1}^{n} z_i z_{n+1-i} t^n = \frac{Z(t)^2}{t},\\
&\sum_{n=1}^{\infty}\sum_{i=1}^{n} z_i t^n = \frac{Z(t)}{1-t},\\
&\sum_{n=1}^{\infty} n t^n = \frac{t}{(1-t)^2}.
\end{aligned}$$

The function equation is

$$\frac{Z'(t)}{t} - \frac{2Z(t)}{t^2} + \frac{1}{t} = \lambda u \frac{Z(t)^2}{t} + 2v\frac{Z(t)}{1-t} + (1-u-v)\frac{t}{(1-t)^2}.$$

Equivalently,

$$Z'(t) = \lambda u Z(t)^2 + \left(\frac{2}{t} + \frac{2vt}{1-t}\right) Z(t) + \frac{(1-u-v)t^2}{(1-t)^2} - 1. \tag{2.2}$$

Substituting $Z(t) = -\dfrac{1}{\lambda u}\dfrac{W'(t)}{W(t)}$ in the convention of a Riccati equation, we get

$$\begin{aligned}
&-\frac{1}{\lambda u}\frac{W''(t)W(t) - W'(t)^2}{W(t)^2} = \frac{1}{\lambda u}\frac{W'(t)^2}{W(t)^2} - \frac{1}{\lambda u}\left(\frac{2}{t} + \frac{2vt}{1-t}\right)\frac{W'(t)}{W(t)} + \frac{(1-u-v)t^2}{(1-t)^2} - 1\\
\iff &-\frac{1}{\lambda u}\frac{W''(t)}{W(t)} = -\frac{1}{\lambda u}\left(\frac{2}{t} + \frac{2vt}{1-t}\right)\frac{W'(t)}{W(t)} + \frac{(1-u-v)t^2}{(1-t)^2} - 1\\
\iff &W''(t) - \left(\frac{2}{t} + \frac{2vt}{1-t}\right)W'(t) + \left(\frac{(1-u-v)t^2}{(1-t)^2} - 1\right)\lambda u W(t) = 0.
\end{aligned}$$

Generating Random Regex Tree Uniformly

May 24, 2026

The goal is to sample recursively, avoid listing all trees. We need to count the number of trees whose root is either $\cdot$, $+$, or $*$ in order to determine the probabilities of choosing each operator at the root. Suppose that all leaves are the same.

3.1 Deriving the Recurrences

Let t_n be the number of regex trees with n leaves that is not a trivial leaf. Let t_n^+ and t_n^* be the number of regex trees with n leaves whose root is $+$, and $*$ respectively. Then

$$t_n = 2t_n^+ + t_n^* \text{ for } n \geq 2, \tag{3.1}$$
$$t_n^* = t_{n-1} \text{ for } n \geq 2, \tag{3.2}$$
$$t_n^+ = \sum_{i=1}^{n-2} t_i t_{n-1-i} \text{ for } n \geq 3, \tag{3.3}$$
$$t_1^+ = t_1^* = 0, t_2^* = 0, t_2^+ = 1, t_1 = 1. \tag{3.4}$$

Using generating functions $T_n(z) = \sum_{k=0}^{n} t_k z^k$ (similarly for T_n^+ and T_n^*), we have

$$\begin{aligned}
T^*(x) &= xT(x) \\
T^+(x) &= x(T(x))^2 \\
T(x) &= x + 2T^+(x) + T^*(x)
\end{aligned}$$

Solving for T, we get

$$T(x) = x + 2(x(T(x))^2) + (xT(x)).$$

We choose the root where $T(0) = 0$.

$$T(x) = \frac{1 - x - \sqrt{1 - 2x - 7x^2}}{4x}. \tag{3.5}$$

Let $\Delta = 1 - 2x - 7x^2$. Then $\Delta' = -2 - 14x$. We have

$$\begin{aligned}
& 4xT(x) = 1 - x - \sqrt{\Delta} \\
\iff & 4T(x) + 4xT'(x) = -1 + \frac{\Delta'}{2\sqrt{\Delta}} \\
\iff & 4T(x) + 4xT'(x) = -1 + \frac{\Delta'\sqrt{\Delta}}{2\Delta} \\
\iff & 8\Delta T(x) + 8\Delta xT'(x) = -2\Delta + \Delta'(4xT(x) - 1 + x) \\
\iff & x\Delta T'(x) + \left(\Delta - \frac{1}{2}x\Delta'\right)T(x) = \frac{(x-1)\Delta' - 2\Delta}{8} \\
\iff & x(1 - 2x - 7x^2)T'(x) + (1 - x)T(x) = 2x.
\end{aligned}$$

Identify the coefficients of x^n with $n \geq 2$ to 0, we get

$$\begin{aligned} & nt_n - 2(n-1)t_{n-1} - 7(n-2)t_{n-2} + t_n - t_{n-1} = 0 \\ \Longleftrightarrow\ & (n+1)t_n = (2n-1)t_{n-1} + 7(n-2)t_{n-2} \end{aligned} \tag{3.6}$$

Given that $t_1 = t_2 = 1$, we can compute t_n in linear time.

3.2 Sampling

We have

$$\sum_{i=1}^{n-2} t_i t_{n-1-i} = t_n^+ = \frac{1}{2}(t_n - t_n^*) = \frac{1}{2}(t_n - t_{n-1}) \text{ for } n \geq 3.$$

The probabilities of the root are

$$p_n(\cdot) = \frac{t_{n-1}}{t_n} \text{ and } p_n(+) = p_n(\cdot) = \frac{t_n^+}{t_n} = \frac{t_n - t_{n-1}}{2t_n}. \tag{3.7}$$

The remaining task is to ensure that the way subtrees are chosen preserves the uniform distribution. We can sample an integer uniformly in $\left\{1, \ldots, \frac{1}{2}(t_n - t_{n-1})\right\}$, regard each term $t_i t_{n-1-i}$ as a block, and determine which block the integer falls into. An optimized way is to order them as $1, n, 2, n-1, \ldots$.

Exact Asymptotic of the Lower Bound

May 30, 2026

4.1 Derivation of the Recurrence

Recall that our shifted yield recurrence is

$$Y_{n+1} \overset{d}{=} I^{(n)} \left(Y_{U_n}^{(1)} + Y_{n-1-U_n}^{(2)} + 1 \right) + J^{(n)} \max \left(Y_{U_n}^{(1)}, Y_{n-1-U_n}^{(2)} \right), \quad Y_1 = Y_2 = 0.$$

Taking expectation on both sides, and letting $y_n = \mathbb{E}[Y_n]$, we have

$$y_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} y_i + \frac{v}{n} \sum_{i=0}^{n-1} \mathbb{E} \left[\max \left(Y_i^{(1)}, Y_{n-1-i}^{(2)} \right) \right] + u, \quad n \geq 1, \quad y_0 = y_1 = 0.$$

Since $\max(Y_i, Y_{n-1-i}) \geq Y_i$ and $\max(Y_i, Y_{n-1-i}) \geq Y_{n-1-i}$, we have

$$\mathbb{E}[\max(Y_i, Y_{n-1-i})] \geq \max(y_i, y_{n-1-i}).$$

Hence, for every $n \geq 1$, we have

$$y_{n+1} \geq \frac{2u}{n} \sum_{i=0}^{n-1} y_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(y_i, y_{n-1-i}) + u.$$

Define (y_n') such that $y_0' = y_1' = 0$ and for every $n \geq 1$,

$$y_{n+1}' = \frac{2u}{n} \sum_{i=0}^{n-1} y_i' + \frac{v}{n} \sum_{i=0}^{n-1} \max(y_i', y_{n-1-i}') + u.$$

Then $y_n \geq y_n'$ for every $n \geq 0$. Finally, we homogenize the recurrence by defining

$$x_n = \frac{2u + v - 1}{u} y_n' + 1$$

for every $n \geq 0$. We have $x_0 = x_1 = 1$ and for every $n \geq 1$. The recurrence of (x_n) is

$$x_{n+1} = \frac{2u}{n} \sum_{i=0}^{n-1} x_i + \frac{v}{n} \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}), \quad n \geq 1, \quad x_0 = x_1 = 1. \tag{4.1}$$

We consider the case where $2u + v > 1$.

4.2 Analysis of the Recurrence

Throughout this section, let α be the unique solution of the equation

$$\alpha + v2^{-\alpha} = 2(u+v) - 1. \tag{4.2}$$

Note that $\alpha > 0$ since otherwise, we would have $2u + v = 1$.

Proposition 1. *The sequence (x_n) is increasing.*

Proof. We have $x_0 = x_1 = 1$ and $x_2 = 2u + v$. Assume that $x_{i+1} \geq x_i$ for all $0 \leq i \leq n-1$ with $n \geq 2$. We have

$$\begin{aligned} nx_{n+1} &= 2u \sum_{i=0}^{n-1} x_i + v \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}) \\ &= 2u \sum_{i=0}^{n-1} x_i + 2v \sum_{i=\lfloor (n+1)/2 \rfloor}^{n-1} x_i + v\delta_n x_{\lfloor n/2 \rfloor}, \end{aligned}$$

where $\delta_n = 1$ if n is odd and $\delta_n = 0$ if n is even.

Let us compute $nx_{n+1} - (n-1)x_n$ for $n \geq 2$. Let $k \geq 1$ be an integer.

- If $n = 2k$, we have

$$2kx_{2k+1} - (2k-1)x_{2k} = 2(u+v)x_{2k-1} - vx_{k-1}.$$

- If $n = 2k+1$, we have

$$(2k+1)x_{2k+2} - 2kx_{2k+1} = 2(u+v)x_{2k} - 2vx_k + vx_k = 2(u+v)x_{2k} - vx_k.$$

Collecting two cases, we have

$$\begin{aligned} & nx_{n+1} - (n-1)x_n = 2(u+v)x_{n-1} - vx_{\lfloor (n-1)/2 \rfloor} \\ \Longleftrightarrow\ & n(x_{n+1} - x_n) = 2(u+v)x_{n-1} - x_n - vx_{\lfloor (n-1)/2 \rfloor} \end{aligned}$$

We have

$$x_n = \frac{2u}{n} \sum_{i=0}^{n-2} x_i + \frac{v}{n} \sum_{i=0}^{n-2} \max(x_i, x_{n-2-i}) \leq (2u+v)x_{n-2}.$$

Hence,

$$\begin{aligned} n(x_{n+1} - x_n) &\geq 2(u+v)x_{n-1} - (2u+v)x_{n-2} - vx_{\lfloor (n-1)/2 \rfloor} \\ &\geq 2(u+v)x_{n-1} - (2u+v)x_{n-1} - vx_{n-1} \\ &\geq 0. \end{aligned}$$

□

Proposition 2. *The sequence (x_n) satisfies the recurrence*

$$nx_{n+1} - (n-1)x_n = 2(u+v)x_{n-1} - vx_{\lfloor (n-1)/2 \rfloor}, \quad n \geq 1, \tag{4.3}$$

with $x_0 = x_1 = 1$.

Lemma 1. *For every $n \geq 0$, we have*

$$M_n = \sum_{i=0}^{n-1} \max(i^\alpha, (n-1-i)^\alpha) \leq 2 \int_{n/2}^{n} x^\alpha dx.$$

Proof. Let $n = 2k$. We have

$$M_{2k} = 2 \sum_{i=k}^{2k-1} i^\alpha \leq 2 \int_k^{2k} x^\alpha dx = 2 \int_{n/2}^{n} x^\alpha dx.$$

Let $n = 2k+1$. We have $k^\alpha = 2 \times \frac{1}{2} \times k^\alpha \leq 2 \int_{k+1/2}^{k+1} x^\alpha dx$. Hence,

$$M_{2k+1} \leq 2 \int_{k+1/2}^{k+1} x^\alpha dx + 2 \int_{k+1}^{2k+1} x^\alpha dx = 2 \int_{k+1/2}^{2k+1} x^\alpha dx = 2 \int_{n/2}^{n} x^\alpha dx.$$

□

Lemma 2. *For every $n \geq 1$, we have $x_n \leq K n^\alpha$ for some $K > 0$.*

Proof. We have $x_2 = 2u + v$. Assume that $x_i \leq K i^\alpha$ for all $1 \leq i \leq n$ with $n \geq 2$ and $K > 0$ is to be determined. For $n \geq 2$, we have

$$\begin{aligned}
n x_{n+1} &= 2u + 2u \sum_{i=1}^{n-1} x_i + v \sum_{i=0}^{n-1} \max(x_i, x_{n-1-i}) \\
&\leq 2u + 2uK \int_1^n x^\alpha dx + 2vK \int_{n/2}^{n} x^\alpha dx \\
&= 2u + \frac{2uK}{\alpha+1} \left(n^{\alpha+1} - 1\right) + \frac{2vK}{\alpha+1} \left(n^{\alpha+1} - (n/2)^{\alpha+1}\right) \\
&= 2u + \frac{K n^{\alpha+1}}{\alpha+1} \left(2u + 2v\left(1 - 2^{-\alpha-1}\right)\right) - \frac{2uK}{\alpha+1} \\
&= 2u + K n^{\alpha+1} - \frac{2uK}{\alpha+1}.
\end{aligned}$$

Note that in the sum of maxima, we do not have to isolate $\max(x_0, x_{n-1})$, since it is automatically evaluated as x_{n-1}, which is already included in the integral. Choose $K = \alpha + 1$, we check that the base cases hold

$$\begin{aligned}
&\alpha + 1 \geq 1, \\
&(\alpha+1)2^\alpha = \left(2(u+v) - v2^{-\alpha}\right) 2^\alpha = 2^{\alpha+1}(u+v) - v \geq 2u + v.
\end{aligned}$$

Therefore,

$$x_{n+1} \leq K n^\alpha \leq K(n+1)^\alpha.$$

□

Theorem 1. *We have $x_n \sim C n^\alpha$ for some $C > 0$ to be determined and α is the unique solution of the equation*

$$\alpha + v2^{-\alpha} = 2(u+v) - 1$$

Proof. Let $z_n = \dfrac{x_n}{(n-1)^\alpha}$ for $n \geq 2$. We will prove that (z_n) converges by proving that $\sum_{n\geq 0} |z_{n+1} - z_n|$ converges. We have, for $n \geq 5$,

$$\begin{aligned} &n^{\alpha+1} z_{n+1} - (n-1)^{\alpha+1} z_n = 2(u+v)(n-2)^\alpha z_{n-1} - v\left(\left\lfloor \frac{n-1}{2} \right\rfloor - 1\right)^\alpha z_{\lfloor (n-1)/2 \rfloor} \\ \iff &z_{n+1} = \left(1 - \frac{1}{n}\right)^{\alpha+1} z_n + \frac{2(u+v)}{n}\left(1 - \frac{2}{n}\right)^\alpha z_{n-1} - \frac{v}{n}\left(\frac{\lfloor (n-1)/2 \rfloor - 1}{n}\right)^\alpha z_{\lfloor (n-1)/2 \rfloor}. \end{aligned}$$

We have

$$\begin{aligned} \left(1 - \frac{1}{n}\right)^{\alpha+1} &= 1 - \frac{\alpha+1}{n} + O\left(\frac{1}{n^2}\right), \\ \frac{1}{n}\left(1 - \frac{2}{n}\right)^\alpha &= \frac{1}{n} + O\left(\frac{1}{n^2}\right), \\ -\frac{1}{n}\left(\frac{\lfloor (n-1)/2 \rfloor - 1}{n}\right)^\alpha &= -\frac{1}{n}\left(2^{-\alpha} + O\left(\frac{1}{n}\right)\right) = -\frac{2^{-\alpha}}{n} + O\left(\frac{1}{n^2}\right). \end{aligned}$$

By the previous lemma, (z_n) is bounded. Therefore,

$$\begin{aligned} z_{n+1} - z_n &= -\frac{\alpha+1}{n} z_n + \frac{2(u+v)}{n} z_{n-1} - \frac{v2^{-\alpha}}{n} z_{\lfloor (n-1)/2 \rfloor} + O\left(\frac{1}{n^2}\right) \\ &= -\frac{2(u+v)}{n}(z_n - z_{n-1}) + \frac{v2^{-\alpha}}{n}(z_n - z_{\lfloor (n-1)/2 \rfloor}) + O\left(\frac{1}{n^2}\right). \end{aligned}$$

Let $\nabla z_n = z_n - z_{n-1}$. Then, there exists a constant $L > 0$ such that for every $n \geq 5$,

$$|\nabla z_{n+1}| \leq \frac{2(u+v)}{n}|\nabla z_n| + \frac{v2^{-\alpha}}{n} \sum_{i=\lfloor (n+1)/2 \rfloor}^{n} |\nabla z_i| + \frac{L}{(n+1)^2}.$$

Equivalently, for every $n \geq 6$,

$$|\nabla z_n| \leq \frac{2(u+v)}{n-1}|\nabla z_{n-1}| + \frac{v2^{-\alpha}}{n-1} \sum_{i=\lfloor n/2 \rfloor}^{n-1} |\nabla z_i| + \frac{L}{n^2}.$$

Let $u_m = \max_{2^m \leq n < 2^{m+1}} |\nabla z_n|$. Then, for every $n \in [2^m, 2^{m+1})$, we have

$$\begin{aligned} \frac{1}{n-1}|\nabla z_{n-1}| &\leq \frac{1}{n-1} \max_{2^m - 1 \leq i < 2^{m+1} - 2} |\nabla z_i| = \frac{1}{n-1} \max(u_{m-1}, u_m), \\ \frac{1}{n-1} \sum_{i=\lfloor n/2 \rfloor}^{n-1} |\nabla z_i| &\leq \frac{n+1}{2(n-1)} \max_{2^{m-1} \leq i < 2^{m+1}} |\nabla z_i| \leq \left(\frac{1}{2} + \frac{1}{n-1}\right) \max(u_{m-1}, u_m). \end{aligned}$$

Therefore,

$$\begin{aligned} |\nabla z_n| &\leq \frac{2(u+v)}{n-1} \max(u_{m-1}, u_m) + v2^{-\alpha}\left(\frac{1}{2} + \frac{1}{n-1}\right) \max(u_{m-1}, u_m) + \frac{L}{2^{2m}} \\ &= \left(\frac{\alpha+1}{n-1} + v2^{-\alpha-1}\right) \max(u_{m-1}, u_m) + \frac{L}{2^{2m}}. \end{aligned}$$

Since $v2^{-\alpha} = 2(u+v) - \alpha - 1 < 1$, we can choose m_0 large enough such that for every $m \geq m_0$ and $n \in [2^m, 2^{m+1})$, there exists γ_1 such that $\dfrac{\alpha+1}{n-1} + v2^{-\alpha-1} < \gamma_1 < \dfrac{1}{2}$. Taking the maximum over $n \in [2^m, 2^{m+1})$, we have

$$u_m \leq \gamma_1 \max(u_{m-1}, u_m) + \frac{L}{2^{2m}}.$$

If $u_{m-1} > u_m$, then

$$u_m \le \gamma_1 u_{m-1} + \frac{L}{2^{2m}} < \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)}.$$

If $u_{m-1} \le u_m$, then

$$u_m \le \frac{L}{2^{2m}(1-\gamma_1)} < \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)}.$$

In both cases, we have

$$\begin{aligned} u_m &\le \gamma_1 u_{m-1} + \frac{L}{2^{2m}(1-\gamma_1)} \\ &\le \sum_{i=m_0+1}^{m} \gamma_1^{m-i} \frac{L}{2^{2i}(1-\gamma_1)} + \gamma_1^{m-m_0} u_{m_0} \\ &< \frac{L}{(1-\gamma_1)} \sum_{i=m_0+1}^{m} \gamma_1^{m-i} \left(\frac{1}{4}\right)^i + \gamma_1^{m-m_0} u_{m_0}. \end{aligned}$$

Choose γ_2 such that $\max\left(\gamma_1, \frac{1}{4}\right) < \gamma_2 < \frac{1}{2}$. We have

$$u_m < \frac{L}{(1-\gamma_1)} \sum_{i=m_0+1}^{m} \gamma_2^m + \gamma_2^{m-m_0} u_{m_0} \le L' m \gamma_2^m,$$

for some constant $L' > 0$. Choose γ_3 such that $\gamma_2 < \gamma_3 < \frac{1}{2}$. Then there exists $m_1 > m_0$ such that for every $m \ge m_1$, we have $L' m \gamma_2^m < L' \gamma_3^m$, implying that,

$$u_m < L' \gamma_3^m.$$

Therefore,

$$\sum_{n=0}^{\infty} |\nabla z_n| = \sum_{m=0}^{\infty} \sum_{n=2^m}^{2^{m+1}-1} |\nabla z_n| = \sum_{m=0}^{\infty} 2^m u_m < \sum_{m=0}^{m_1-1} 2^m u_m + L' \sum_{m=m_1}^{\infty} (2\gamma_3)^m < \infty.$$

Hence, (z_n) converges to some constant $C > 0$. Thus, $x_n \sim C n^{\alpha}$. $\square$

References

[1] Florent Koechlin and Pablo Rotondo. “The effects of semantic simplifications on random BST-like expression-trees”. In: *Discrete Mathematics* 347.5 (2024), p. 113906.